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Purposefully different,

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Welcome to EEN1037: Web Applications Development

Introduction to the Web

Week 1

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What will I learn?

- What is the Web?
- How is the Web built?
- What is W3C and why it is so important?
- What are the HTTP, FTP and HTTPS?
- What is a URI, URL, IP, and DNS?
- What is HTML and where to find it?

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The Web: what is it?

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The Web: once upon a time…

Vannevar Bush, As We May Think, the “Memex” (1945)

Ted Nelson, Hypertext - Xanadu project (1965)

Tim Berners-Lee, ENQUIRE (1990)
The World Wide Web (1989), CERN

1st Web page (late edition 1992)
http://www.w3.org/History/19921103-hypertext/hypertext/WWW/TheProject.html

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The Web (= Internet)

- The whole set of resources (text, audio, video)
  - Connected via (Hyper) links
  - Available on the Internet
  - Provided by *servers*
  - Accessible by *clients* via Browsers
    - Chrome, Firefox, Internet Explorer, Safari …
  - Created by
    - Organization
    - Enterprises
    - People (Web 2.0)

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The current Web

- Not just hyper-text linking resources
- The “biggest database” ever built by people
- Mobile devices and connectivity
- A lot of information is hidden in the “deep web” (dynamic data, behind forms, services etc.) not accessible to search engines – approx. 500 times bigger than the “surface web”!!
- Web concepts used “off-line” – inside organizations (intranet)
- User-generated content – large increase in the amount of data
- New possibilities, but also information overload
- http://milliondollarhomepage.com/

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Web and its Science: Measuring the Web

- Importance of gathering statistics about the web and understand its properties
- Questions:
  - Which websites are more important?
  - Which communities are more active?
  - Which source is more reliable?
  - How big is the web? http://www.worldwidewebsize.com

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World Wide Web Consortium (W3C)

- WWW/The Web Conference:
  - First International Conference on the World-Wide Web May 1994
  - First W3 Consortium Meeting: Dec 1994
- W3C Consortium : http://www.w3.org
- Core Principles:
  - Organized in Working Groups, Members + experts
  - Specifications are made available since early versions
  - Consensus
  - Strict licensing and Patenting policies
- Not only the Web (HTML) but also many important standards: XML (XML 1.0 Recommendation published on 10th February 1998), XML Schema,....

What’s under the hood?

- Web essential ingredients
- Web architecture

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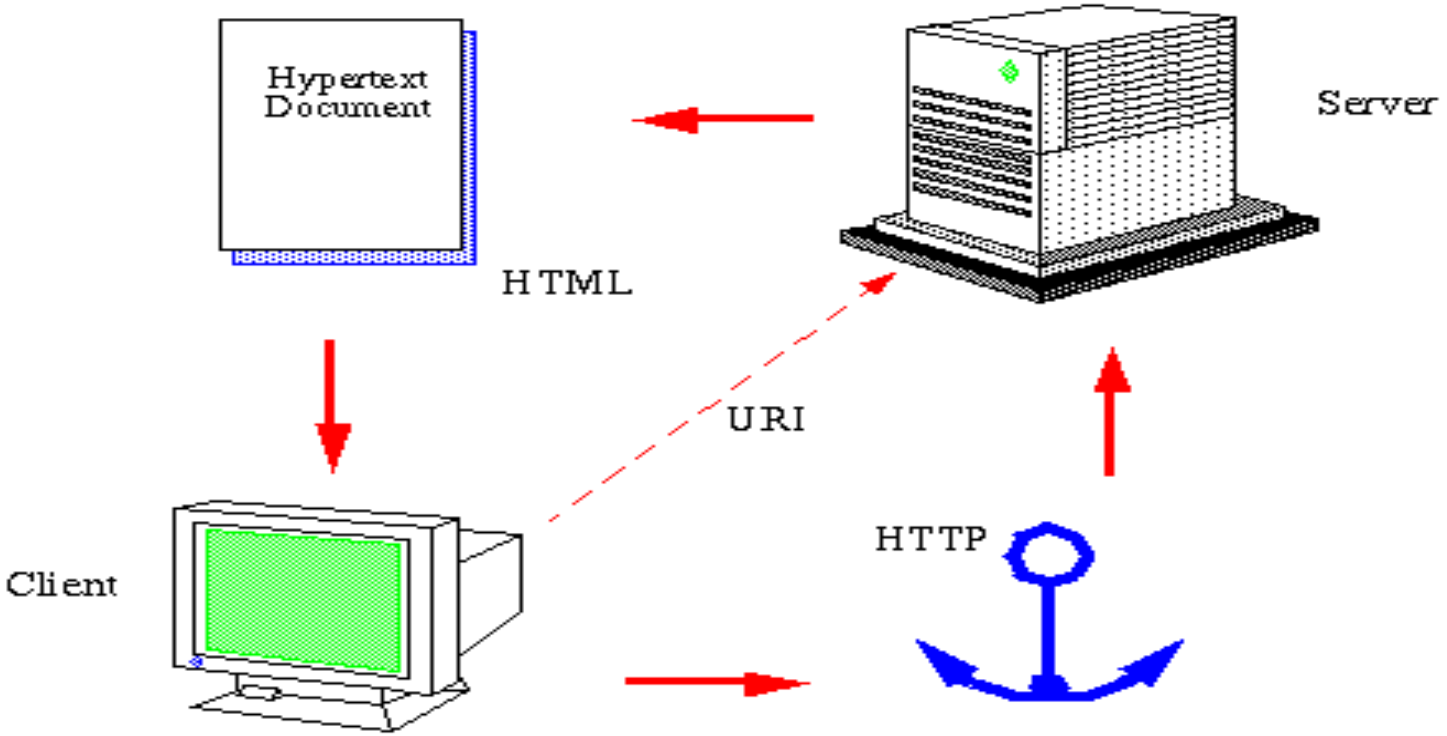
The Web Architecture



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Essential ingredients

- Hosting Server – the computer that is connected to the web using appropriate protocols and stores website files in order to make them available to requesting entities.
- Web client – program (browser) that requests, receives and displays web objects (HTML pages, pictures, audio, …)
- Protocol – convention all computer programs use to communicate something with each other
  - HTTP – Hypertext Transfer Protocol – set of rules to enable web browsers to communicate with web servers
  - IP – Internet Protocol (it relies on the IP address of a resource, e.g. 140.203.201.74)
  - Other important protocols:
    - FTP – File Transfer Protocol (see ftp://ftp.microsoft.com/)
    - SMTP/IMAP – Simple Mail Transfer Protocol

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Essential ingredients

- HTML – Hypertext Markup Language

- Uniform Resource Locator (URL) – web address that explains how to find the website
  - http://dcu.ie/
- Domain Name Server (DNS) – special server that plays the role of a telephone book – translates the URL into a numerical, physical Internet Protocol (IP) address

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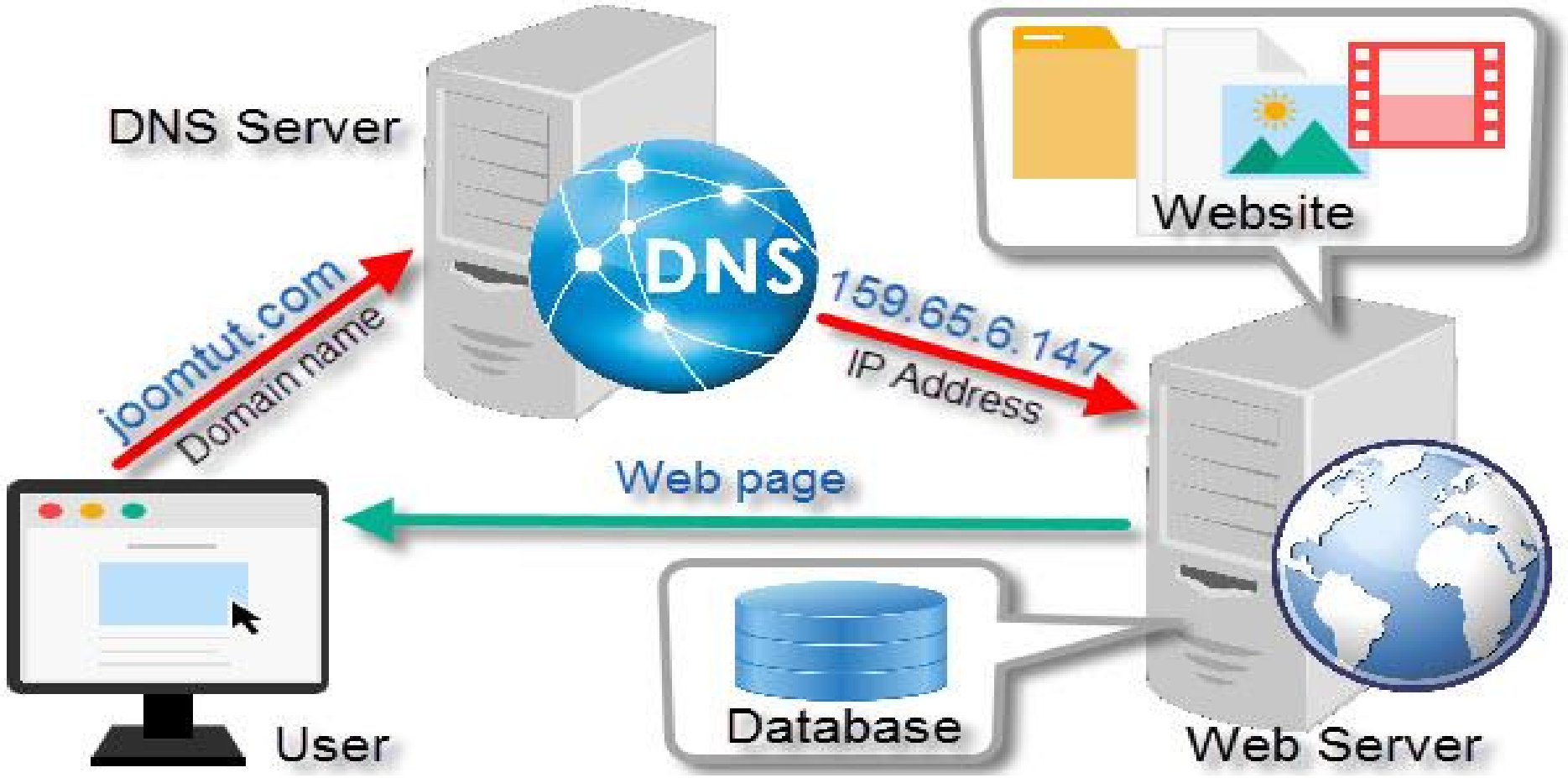
A simplistic view of Web information flow



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Hyperlinks

- A hyperlink (or link) is a word, group of words, or image that you can click on to
  - jump to a new document or
  - jump to a new section within the current document.
- When you move the cursor over a link in a Web page, the arrow should turn into a little hand to indicate the linkage.
- Source: http://w3schools.com/html/html_links.asp

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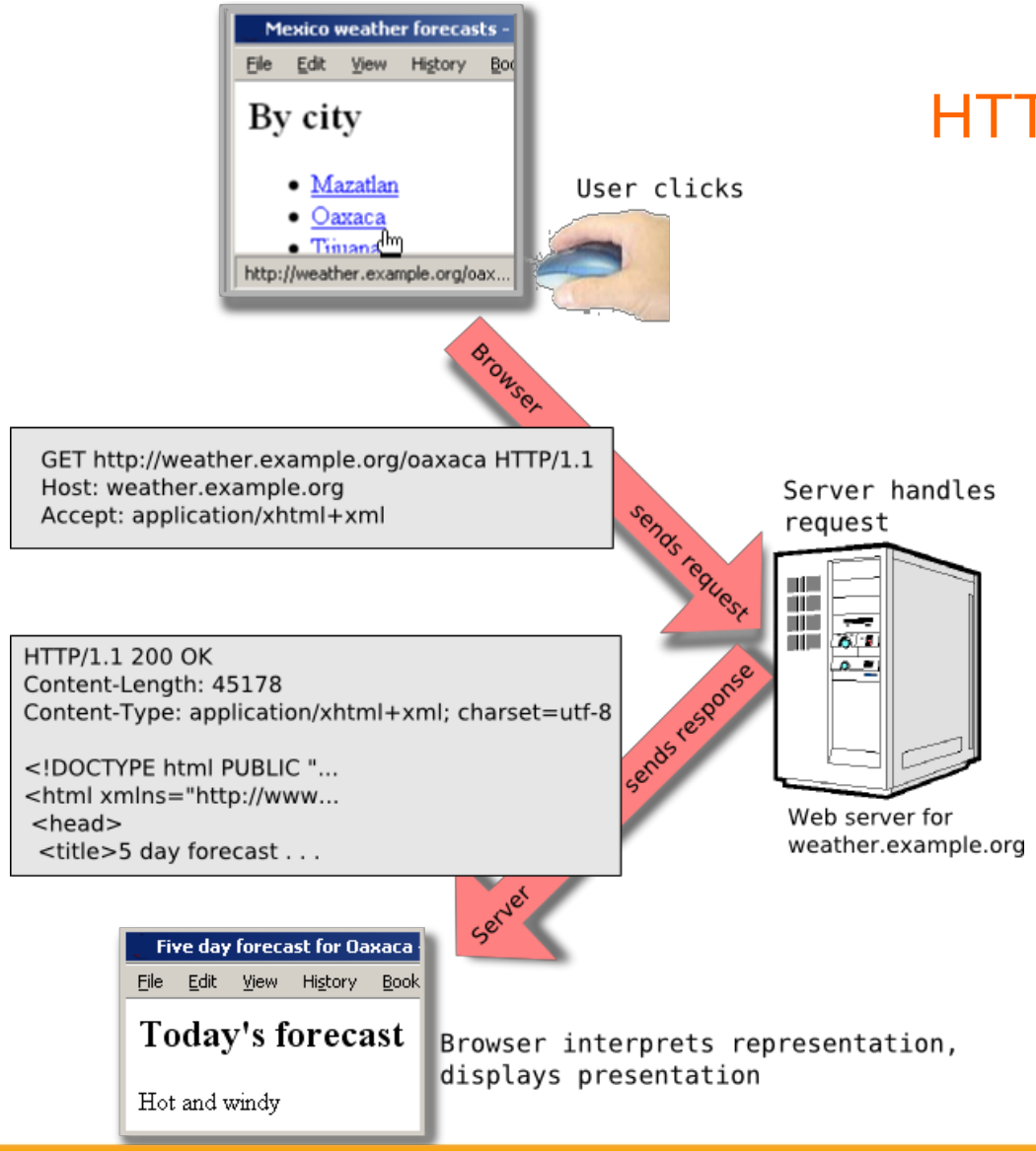
When I click the hyperlink…

HTTP Protocol



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Request

- Sent from the browser to the server
- Contains the request line and the header lines
  - Request line: what operation we would like to execute on a given URL and using which protocol
  - Header lines: additional information, e.g. type of the accepted content

Request line → **GET http://www.example.ie/somepage.html HTTP/1.1**

Header lines

Host: example.ie

Accept: application/xhtml+xml

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Response

- Contains the status line, header lines, and (potentially) data
  - Status line informs the browser if the request was successful
  - Header lines provide details of the request and content

Status line (protocol and status)

Header lines

Data, e.g. requested HTML

HTTP/1.1 200 OK
Date: Mon, 07 Dec 2020 12:00:15 GMT
Server: Apache/1.3.0 (Unix)
Last-Modified: Mon, 22 Jun 2020 .....
Content-Length: 6821
Content-Type: text/html

<!DOCTYPE public ...

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Let’s see it live (at least once)

- Run Google Chrome
  - https://www.google.ie/chrome/browser/desktop/index.html
  - Install HTTP headers Chrome Extension:
  - Visit http://www.dcu.ie/

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